

PD Interview Questions

- Q.1 Insertion delay will affect on frequency? How? What is the relation between Insertion delay and frequency?
- Q.2 On what factors cell delay depends on?
- Q.3 what are calibre Drc's you have seen which are not shown in Innovus
- Q.4 How is clock uncertainty calculated in your flow? Based on which all parameters ?
- Q.5 why we need softchecks, ERC checks and PERC checks in LVS
- Q.6 If you have 40 corners in your design? Do you need to fix violations in all 40 corners?
- Q.7 What happens if i use type of clock buffer which is bad rise time and good fall time?
- Q.8 How to decide optimum H & W of core, given Aspect Ratio & utilization? Implications if $H > W$ & vice- versa?
- Q.9 draw stick diagram of nand gate
- Q.10 If you have a 1000 macros how will start floorplanning.
- Q.11 Why is clock skew constraint specified as greater than 0 ps or a specific value (30 - 50 ps) ? Why not it is 0 ?
- Q.12 Fixes to Antenna violations: Layer Hoping (jumper/Bridging) and diode insertion (reversebiased), upsize gate (increase gate area).
- Q.13 what are the components will you see in timing report
- Q.14 Could you please draw internal structure of cmos Nand, nor, isolation, retention, level shifter, end cap , D cap , tap cells etc....
- Q.15 In TSMC 7nm technology which track cells you have used and what is the poly width(gate length)
- Q.16 cell delays or net delays which one will be more in lower technologies ? why ?

- Q.17 how much cap. And tran values u shud specify in the design
- Q.18 Inverter characteristics? Why PMOS will be on at logic 0, and NMOS will be on at logic 1?
- Q.19 What we do in placement? What we do if congestion exists ? and regarding timing we do setup analysis after placement...
- Q.20 Why CMOS consumes less power?
- Q.21 What are the uses of clock gate. How many types of clock gate will be available in design
- Q.22 Explain effects of clock domain crossing?
- Q.23 How do you minimize clock dynamic power ?
- Q.24 Which clock tree is better, the one with shielding on a single spacing or the one with double spacing?
- Q.25 Zero cycle?? Was talking regarding Input to flop paths..
- Q.26 What will be the impact on hold time if your skew is more??
- Q.27 If design meets 300MHz, does it meet 1GHz?
- Q.28 After routing if you have 1000 unique setup violations? how we will fix? can we fix with tool? How?
- Q.29 command for adding physical cells like end cap/ tap/ filler etc...
- Q.30 Design Frequency will affect the cross talk? If yes... why
- Q.31 What is difference between high drive strength buffer and low drive strength buffer?
- Q.32 How shielding exactly reduces crosstalk?
- Q.33 What is the clock gate..? explain with diagram and operation
- Q.34 What is ICG and its pros and cons?
- Q.35 What is IR drop? How it will effect to design?

- Q.36 Which type of violations will get .. due to cross talk
- Q.37 when -combo switch is used in create_generated_clock
- Q.38 What is signal integrity? what happens if design have signal integrity problems?
- Q.39 why and in what cases false path is applied ? lly disable timing arc ?
- Q.40 in CTS what is preferred first to fix .. logical drcs , skew ,ID
- Q.41 Why is that you see new violations after Clock tree synthesis? Do the clocks have to be balanced for the spare cells, explain?
- Q.42 Timing window concept?
- Q.43 What is the cross talk effect between 1MHz design to 1GHz design
- Q.44 How do unconnected (or) floating Input ports impact PD flow?
- Q.45 How will you know that two clocks are synchronous?
- Q.46 What are the inputs for running LEC? What are the issues faced in LEC?
- Q.47 What the difference between synthesis and High Fan-out synthesis
- Q.48 why inverter is preferred in CTS over buffer
- Q.49 what are the inputs to run STA? faced any issues in extraction?
- Q.50 Is Hold dependent on frequency? Explain in detail?
- Q.51 What is the reason behind that, if the temperature increases why delay increasing?
- Q.52 Switching transitions effects the cross talk ?
- Q.53 What is transition violation? How it effects on your design power (In terms of short circuit power)?
- Q.54 What will happen, when Victim is at constant high, Aggressor changing from logic 0 to logic 1.

- Q.55 Will P & R tool reject synthesis netlist with unconnected Input ports?
- Q.56 inputs for crosstalk analysis?
- Q.57 What is temperature inversion?
- Q.58 How to find the reason for congestion in particular region?
- Q.59 How will cell sizing affect dynamic, leakage and short circuit power?
- Q.60 Have you done any scripting for your designs?? About db commands..
- Q.61 Which is preferred in CTS: clock buffer or clk inverter?
- Q.62 How can you write script to fix 1000 setup unique violations?
- Q.63 How we will balance the multiple clocks in a design?
- Q.64 How power stripes are useful in power planning? What are reliability issues?
- Q.65 Explain challenges in previous design
- Q.66 Can we build clock tree without giving skew and insertion delay constraints?
- Q.67 What was delay of the buffer that you used in your project?
- Q.68 Name the types of ECO in P&R. Please provide a flow to for the different type of ECO.
- Q.69 If typical is 1 volt , then what will be the corners for worst and best and why ?
- Q.70 Timing windows effect on xtalk noise.
- Q.71 In which stage SDC is required? Can you place design without SDC?
- Q.72 what is logically exclusive and physically exclusive.
- Q.73 How extraction is taking OCV & crosstalk effects? How timing is taking both these effects?
- Q.74 Is mix Vt possible in CTS? Why? Why one VT cells are preferred in CTS?
- Q.75 What are the corners used in previous projects and explain how they work.

Q.76 If you and your team are working in new technology(ex: 5nm/3nm), what are the ways to escalate issues

Q.77 What are the multicycle paths? How we check Setup and Hold for multicycle path?

Q.78 How buffer insertion to a victim net reduces crosstalk effects?

Q.79 How do you prevent transition violations

Q.80 What is synthesis and explain why it requires?

Q.81 If skew is violated, how will you fix it?

Q.82 Pulse width violations: How is it imp & fixes?

Q.83 how will u say constraints u r getting from frontend are ok

Q.84 Schematic diagram and truth table on Isolation cells and retention flops. And switch cell schematic structure

Q.85 Launch path with Clock - 2ns period, Insertion delay - 300ps, clk2q delay - 30ps, combo delay - 1.2ns, Tsetup with -10ps, Setup uncertainty -50ps, Capture Path with Clock - 3ns period, Insertion delay - 300 ps, Will it meet setup and Hold ? If not why ? similarly try with different clock periods and insertion delays and check will you meet hold and setup ?.

Q.86 How will you conclude a path as multicycle path??

Q.87 DW-DS for CLK Non -default Routing? Why?

Q.88 Xtalk noise on static signals and glitch effects on static signals.

Q.89 what is the purpose of via enclosure

Q.90 What will happen will cross talk goes high

Q.91 How to come up with CTS specs: max delay values & max skew values?

Q.92 how clock buffer different from normal buffer

Q.93 What is difference between physical DRC violations and timing DRC violations?

Q.94 what happens if duty cycle is ON period 30% and OFF period 70% instead of 50% duty cycle?

Q.95 what are the disadvantages of having good skew.

Q.96 What are the inputs for Prime Time?? About Prime time(STA) analysis and reports generated? How do you resolve this issue reported in prime time ??

Q.97 There is path from the launch clock at 2ns and capture clock at 3ns, where the tool checks the setup and hold? Write the command for hold check in PT?

Q.98 If two buffers are added in the common path (timing path), then what will be the effect on hold in PT? SI soft check errors in LVS

Q.99 What is uniquifying the netlist.

Q.100 What is the amount of input and output delays that are used? For the 70% 30% approach write down the timing report of the In to Out paths with Data Arrival and Data Required times, for a 100MHz clock and 2ns combinational delay.

Q.101 How to solve Electro migration violations?

Q.102 How much congestion metric is good to go

Q.103 Can you close the design without Clock tree (I'm not bothering timing)

Q.104 While comparing 1MHz to 1GHz, which design having more hold violations ? why..?

Q.105 Why we need to do Cell padding. How it will reduce congestion.

Q.106 Why via sizes are increasing in lower technologies?

Q.107 No. of levels in CTS , can we specify in the clocktree file

Q.108 What are the pros and cons of up sizing and downsizing a cell?

Q.109 Where do you find the DRV values? how do you analyse in the timing report?

Q.110 Role of following cells in latest technologies: Endcap/Endrow cells, decaps, Core fillers, metal fillers, tie-hi/tie-lo cells, Well tap cells?

Q.111 How to decide the placement was good?

Q.112 what are dpt layers in your design and why will go for two masks for those dpt layers

Q.113 Why is the term lockup latch is used explain it by developing a scenario.

Q.114 In sdc what is synchronous and asynchronous clock groups

Q.115 write a tcl script to get the number of AND gates in the fanin cone of D pin of register

Q.116 What is the command you will use for adding stripes?

Q.117 Can you build CTS without SDC? If it is not why? Without SDC can we close the design?

Q.118 If spacing is not recommended in your design, what are the other techniques you perform to avoid CC?

Q.119 What are the checks you do after the placement?